MRI Image Classification for Alzheimer's Diagnosis: Development and Evaluation of a Custom CNN Model

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***Abstract*—This study presents the implementation and evaluation of a custom Convolutional Neural Network (CNN) model designed for the classification of brain MRI images in the diagnosis of Alzheimer's disease. The dataset, comprising multiple classes of Alzheimer's images, was divided into training, validation, and test sets. Two versions of the model were developed: one using data transformation and one without. Within each version, the model's performance was further evaluated by experimenting with different subsampling techniques (MaxPooling vs. Stride 2) and regularization methods (with vs. without dropout). The CNN architecture consists of multiple convolutional blocks with batch normalization, ReLU activation, and the aforementioned optional layers. The model was trained using class-weighted loss functions to address class imbalance and optimize performance. The results indicate the model's effectiveness in differentiating between various stages of Alzheimer's, with detailed performance metrics such as accuracy, precision, recall, and confusion matrix analysis provided. Additionally, an early stopping policy was implemented to avoid overfitting during training.**

I. Introduction

Alzheimer's disease (AD) is the most common cause of dementia globally, responsible for 60-80% of cases [1]. Characterized by the accumulation and deposition of cerebral amyloid-β (Aβ), AD steadily worsens from mild to severe, impairing the ability to complete any task without assistance. The prevalence of AD is closely correlated with age, with a diagnosis rate of less than 1% for individuals aged 60-64, escalating to 20-40% for those over 85 years of age [2]. Currently, more than 55 million people are affected worldwide, with the majority residing in low- and middle-income countries. With nearly 10 million new cases each year, AD is the seventh leading cause of mortality and significantly contributes to disability and dependence among the elderly [3].

In the field of Alzheimer's disease diagnostics, Magnetic Resonance Imaging (MRI) is emerging as a highly effective methodology. MRI, known for its ability to capture detailed anatomical information, enables clear visualization of brain structures, aiding in the early diagnosis of Alzheimer's-related manifestations. It provides precise data on the physical composition and integrity of the brain, helping recognize critical signs of AD development, such as brain shrinkage, ventricular enlargement, and hippocampus volume reduction. Furthermore, MRI scans help distinguish AD from other neurodegenerative disorders and assess disease progression over time, facilitating timely interventions and improving diagnostic accuracy.

The paradigm shift introduced by deep learning, specifically Convolutional Neural Networks (CNNs), has revolutionized the landscape of medical image classification, including Alzheimer's diagnostics. CNNs autonomously extract features directly from image data, eliminating the need for manual feature extraction and yielding exceptional results in accuracy and performance. By employing convolutional filters, CNNs capture intricate patterns and features across different scales and levels of abstraction within Alzheimer's-related MRI scans. The incorporation of pooling layers and fully connected layers further empowers the creation of adaptable and scalable models, overcoming the limitations of conventional approaches.

In this study, a customized CNN framework is proposed to detect specific features of Alzheimer's disease from MRI images. The experiment is performed on a public dataset titled "*Alzheimer MRI Preprocessed Dataset*" obtained from Kaggle. This dataset is widely recognized within the scientific community and includes data collected from various websites, hospitals, and public repositories.

The investigation evaluates the model's performance under two primary configurations: one where various data transformations are applied to the images, and another where the images remain in their original state. Additionally, in each configuration, the study examines the effect of the presence or absence of Dropout for regularization and compares the impact of MaxPooling with that of Stride 2. This approach allows for a comprehensive analysis of how these techniques influence the model's ability to classify Alzheimer's-related features in MRI scans.

By comparing the performance on the transformed and non-transformed datasets, this study aims to assess the model's robustness and generalizability under both ideal conditions (non-transformed images) and more realistic scenarios where images may be subject to various distortions and blurring.

II. Related Work

In recent years, deep learning has gained considerable interest for its use in the diagnosis of Alzheimer's disease. Various deep learning methods have recently been introduced as diagnostic tools for Alzheimer's, supporting physicians in making informed clinical decisions. In this section, some studies that that have contributed to the state-of-the-art in this domain will be presented.

Lu et al.'s article [4] from 2018, introduces a novel deep learning framework for the early diagnosis of Alzheimer's disease using multimodal, multiscale deep neural networks.

This method exploits both structural MRI and FDG-PET images to improve accuracy in classifying individuals with mild cognitive impairment (MCI) progressing to Alzheimer's. Their method highlights the critical role of combining complementary data from various imaging techniques for a more accurate diagnosis. This approach achieved an 82.4% accuracy in predicting which individuals with mild cognitive impairment (MCI) would progress to Alzheimer's disease (AD) within three years. Additionally, the sensitivity in identifying individuals with a probable AD diagnosis was 94.23%, while it reached a combined accuracy of 86.3% for the non-demented class, outperforming results reported in the existing literature.

In their 2019 article, Gupta et al. [5] proposed a diagnostic method for Alzheimer's disease (AD) classification by comparing the effectiveness of combined versus individual features. They employed three traditional machine learning classifiers: K-nearest neighbor (KNN), random forest (RF), and support vector machine (SVM). The experimental results demonstrated that the integration of features derived from three techniques—Voxel-Based Morphometry, Cortical and Subcortical Segmented Region, and Hippocampal Volume—yielded superior performance in classifying AD or mild cognitive impairment (MCI) compared to using individual features. Notably, the support vector machine (SVM) classifier achieved the highest accuracy of 96.42% in distinguishing AD from Healthy Control (HC) subjects.

Ahmed et al. [6] proposed a model for diagnosing Alzheimer's disease using structural magnetic resonance imaging (sMRI). To improve diagnostic accuracy and prevent overfitting, the model employs ensembles of simple convolutional neural networks (CNNs) as feature extractors and uses softmax cross-entropy as a classifier. Utilizing a patch-based approach due to limited data, the model achieved 90.05% accuracy, comparable to state-of-the-art methods, with only sMRI data of the hippocampus.

Nawaz et al. [7] in their article from 2021 proposed a system for detecting the stages of Alzheimer's disease (AD) using a pre-trained AlexNet model to address class imbalance. The model uses the initial layers of AlexNet to extract deep features, which are then classified using SVM, KNN, and Random Forest, achieving a superior accuracy of 99.21%, surpassing existing state-of-the-art methods.

Afzal et al. [8] in their article addressed the challenge of unbalanced training data by proposing a framework that uses data augmentation and transfer learning with AlexNet. They augment 3D MRI images and achieve high accuracies: 98.41% with a single MRI view and 95.11% with 3D views, outperforming existing methods for AD stage classification. Their combined approach effectively tackles data imbalance and enhances classification accuracy for AD stages.

Murugan et al. [9] presented DEMentia NETwork (DEMNET), a model employing an efficient convolutional neural network (CNN) to detect AD-specific features from MRI images, while addressing class imbalance using the SMOTE technique. With an accuracy of 95.23%, DEMNET outperforms existing methods effectively identifying brain regions associated with AD and serving as a decision support system for predicting disease severity. Future work aims to enhance DEMNET's performance through training on diverse datasets, establishing it as an autonomous framework for dementia screening and AD diagnosis.

Khanna [10]'s study from 2023 employs MobileNetV2 and LSTM through deep ensemble learning (DEL) on MRI images from the ADNI dataset to classify dementia stages. DEL outperforms traditional CNN approaches, achieving notable metrics such as 94% sensitivity, 95% specificity, and superior performance across accuracy, AUC, F1-score, precision, recall, and computational efficiency.

In their research, Singh and Kumar [11] utilise many pre-trained Convolutional Neural Network (CNN) models like EfficientNet, MobileNet, DenseNet, ResNet, AlexNet, InceptionV2, and NASNet to classify six stages of Alzheimer's disease (AD): Normal Control (NC), Significant Memory Concern (SMC), Early Mild Cognitive Impairment (EMCI), Late Mild Cognitive Impairment (LMCI), and AD. The models achieved average accuracies of 99.79%, 98.74%, 84.39%, 84.28%, 83.74%, 94.39%, and 88.28%, respectively. Evaluation included recall, AUC, loss values, F1 score, and precision metrics. Rigorous preprocessing of raw data was undertaken to enhance classification accuracy, demonstrating the efficacy of deep learning models in AD diagnosis.

In their recent article (2024), Arafa et al. [12] highlight the importance of early AD diagnosis and introduce a deep learning framework using CNNs for medical image classification. This framework includes data preprocessing, augmentation, cross-validation, and classification, utilizing both a simple CNN and a pre-trained VGG16 model with transfer learning and fine-tuning. Data augmentation techniques (such as rotation, translation, and noise addition) were employed to address class imbalance and prevent overfitting. The proposed models achieved high accuracies, with 99.95% and 99.99% for the CNN and 97.44% for the VGG16 model.

The literature shows that there are numerous techniques for classifying Alzheimer's disease (AD) using machine and deep learning. However, the problem of high model parameters and class imbalance in multiclass AD classification remains a challenge. A CNN custom model with a reduced number of parameters is proposed below to address this problem. Furthermore, the technique of class weights was employed to resolve the class imbalance in the data, allowing the four stages of dementia that can lead to AD to be accurately identified and predicted.

III. Proposed Approach

A. Dataset and Preprocessing

As briefly mentioned in the I section, the dataset that will be used is the *Alzheimer MRI Preprocessed Dataset* [13], which consists of 6400 MRI Images of four classes: *Mild Demented* (896 images), *Moderate Demented* (64 images), *Non-Demented* (3200 images), and *Very Mild Demented* (2240 images). All the images are already resized into 128 x 128 pixels. The library used to realise the project is PyTorch [14], as it offers an excellent balance between usability and powerful expressive capabilities.

The preprocessing of the dataset began with converting the images to tensors, and normalizing pixel values to a range of

0 to 1 in order to improve the performance of neural networks by standardizing input values.

Then the dataset was divided so that the 80% percent was designated for the training set, 10% percent for the validation set, and the remaining 10% for the test set. This division ensures that a substantial portion of the data is used for training the models, while separate sets are reserved for validation and rigorous evaluation.

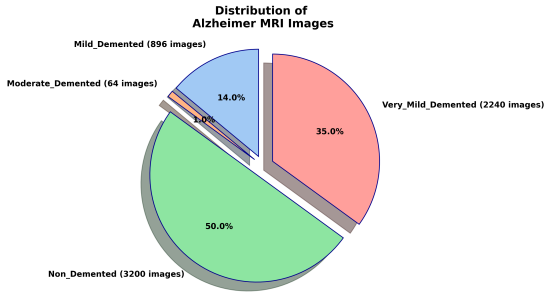


Fig. 1: Pie Chart of class distribution showing a high imbalance

Since the dataset exhibits a high class imbalance, as shown in Fig. 1, class weighting technique was applied. Class weights were calculated to balance the training process by assigning higher weights to the minority classes. These weights were then used in the definition of the loss function to ensure fair representation during the optimization process, helping to reduce knowledge loss and minimize overfitting without increasing the original dataset size.

The final step in the dataset preprocessing involved applying data transformation techniques to enhance training data diversity, performed solely in one notebook. Transformations included random rotation (up to 30 degrees), random resized cropping (to 128 pixels with a scale of 0.8 to 1.0), Gaussian blur (kernel size 3), brightness adjustment (by 0.2), contrast adjustment (by 0.2), and hue adjustment (by 0.01). These techniques aimed to broaden the model's exposure to varied inputs, thereby improving generalization and prediction accuracy on unseen data.

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B. Model Architecture

As previously noted, the study explores various configurations across two separate notebooks, each evaluating a convolutional neural network (CNN) under different conditions: with and without data transformations.

In both configurations, the batch size is fixed at 64.

Additionally, within each notebook, the following choices are considered:

- using Stride 2 versus Max Pooling;
- including a Dropout layer or not

This setup results in eight distinct configurations (four with data transformations and four without) for comparative analysis.

The proposed architecture, as illustrated in Fig. 2 and Fig. 3, consists of several key components designed for extracting features from the input images. The architecture includes 7 convolutional blocks, each followed by *batch normalization* layers, and *ReLU* activation functions. Additionally, the architecture includes an optional *Max Pooling layer* or an alternative use of *stride 2* in the convolutional layers to achieve down-sampling.

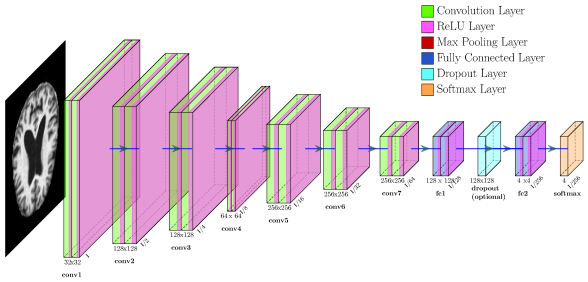


Fig. 2: Graphical representation of the architecture when Stride 2 is chosen (Drop Out is optional)

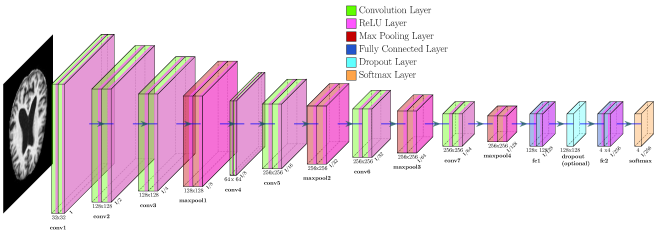


Fig. 3: Graphical representation of the architecture when MaxPooling is chosen (Drop Out is optional)

The model is created with the ability to toggle between using dropout and max pooling. This flexibility allows for experimentation with different configurations to observe their effects on model performance.

1) The input, with dimensions $128 \times 128 \times 3$, enters the first 2D convolutional layer comprising 32 filters of size 3×3. This layer applies a padding of 1 and a stride of 1 to ensure that the spatial dimensions are preserved. The output then passes through a batch normalization layer and a ReLU activation function.
2) The second convolutional block consists of a 2D convolutional layer with 128 filters of size 3×3, padding of 1, and stride of 1. This is followed by batch normalization and a ReLU activation function.
3) The third convolutional block includes a 2D convolutional layer with 128 filters of size 3×3, padding of 1, and a stride of 1 if max pooling is used, otherwise stride is set to 2. After batch normalization and ReLU activation, max pooling with a 2×2 window and stride of 2 is optionally applied.
4) The fourth block features a 2D convolutional layer with 64 filters of size 1×1, padding of 0, and stride of 1. This layer acts as a 1x1 pooling layer, used to reduce the

dimensionality of the channels while keeping the spatial dimensions of the input unchanged. This is followed by batch normalization and ReLU activation.

5) The fifth convolutional block includes a 2D convolutional layer with 256 filters of size 3×3, padding of 1, and a stride of 1 if max pooling is used, otherwise stride is set to 2. Batch normalization and ReLU activation are followed by optional max pooling.
6) The sixth convolutional block consists of a 2D convolutional layer with 256 filters of size 3×3, padding of 1, and a stride of 1 if max pooling is used, otherwise stride is set to 2. Batch normalization and ReLU activation are followed by optional max pooling.
7) The seventh convolutional block includes a 2D convolutional layer with 256 filters of size 3×3, padding of 1, and a stride of 1 if max pooling is used, otherwise stride is set to 2. Batch normalization and ReLU activation are followed by optional max pooling.
8) After the final convolutional block, the output feature maps are flattened into a one-dimensional tensor. The flattened output is then fed into a fully connected layer with 128 neurons. This layer is followed by an optional dropout layer (i.e. if dropout is selected) with a dropout rate of 0.5, and a ReLU activation function.
9) Finally, the output passes through a fully connected layer with 4 neurons, corresponding to the number of classes in the classification problem. The final activation function is a log softmax, providing the log-probabilities of each class.

The training process employs the *Adam optimizer* with a learning rate of 0.0001, and the loss function used is *CrossEntropyLoss*, adjusted with class weights to address any class imbalance, as previously mentioned. Additionally, a learning rate scheduler, *ReduceLROnPlateau*, is implemented to dynamically reduce the learning rate when a metric has stopped improving, thus aiding in better convergence.

To evaluate the model's performance, metrics such as accuracy, precision, recall, and F1-score are used. These metrics provide a comprehensive assessment of the model's ability to correctly classify images across different classes.

The adoption of a batch normalization layer made the inclusion of dropouts in the convolutional part of the network unnecessary. Generally, the combined use of these techniques is discouraged since, as a rule of thumb, it tends to lead to worse results than using only one of them.

IV. Experiments

Experiments were conducted on several parameters of the previously described network and training components. These included the method used for subsampling the dimensions of the convolutional filter responses (Stride 2 versus Max Pooling) and the optional inclusion of dropout layers. The training was carried out for 20 epochs. To ensure reproducibility of the results, a fixed random seed of 42 was used.

To prevent overfitting and ensure the model's best performance, *early stopping* is employed during this training process. Early stopping monitors the validation accuracy, and if there is no improvement for a specified number of epochs (patience, here is 5), the training process is halted. This prevents the model from continuing to train when further improvements are unlikely, thus saving computational resources and reducing the risk of overfitting.

A. Training with Data Transformation

The training of the network on the transformed dataset yielded the results shown in the table I, divided according to the parameter combination used.

	Stride (2,2)	**MaxPooling**
No Drop out	97.50%	96.09%
Drop out	97.03%	96.56%

TABLE I: Classification accuracy on test set with data transformation

The results indicate that using a stride of (2,2) for subsampling without dropout layers achieves the highest accuracy of 97.50% on the transformed dataset. In comparison, the inclusion of dropout layers slightly reduces the accuracy to 97.03%.

When max pooling is used instead of stride (2,2), the highest accuracy achieved is 96.56% with dropout layers, which is lower than the corresponding configuration with stride (2,2). Notably, max pooling without dropout layers results in the lowest accuracy of 96.09%.

Overall, these results suggest that using stride (2,2) for subsampling is more effective than max pooling in the context of data transformation. Additionally, the inclusion of dropout layers does not provide a significant improvement in performance and can slightly reduce the accuracy in some cases.

B. Training without Data Transformation

The training of the network on the original dataset, without any transformation, yielded the results shown in the table II, divided according to the parameter combination used.

	Stride (2,2)	**MaxPooling**
No Drop out	98.13%	96.25%
Drop out	98.28%	95.63%

TABLE II: Classification accuracy on test set without data transformation

In the absence of data transformation, using a stride of (2,2) for subsampling with dropout layers achieves the highest accuracy of 98.28%. This is a slight improvement over the configuration without dropout layers, which achieves an accuracy of 98.13%.

On the other hand, when max pooling is used, the highest accuracy achieved is 96.25% without dropout layers. The inclusion of dropout layers in this case leads to a further reduction in accuracy to 95.63%.

These results suggest that, without data transformation, the use of dropout layers can provide a slight improvement in performance when stride (2,2) is used for subsampling. However, max pooling generally results in lower accuracies compared to stride (2,2), regardless of the presence of dropout layers.

C. General considerations

From the results presented in the previous tables concerning model training with and without data transformation, several general conclusions can be drawn regarding the effectiveness of stride, max pooling, and dropout parameters, and their impact across different data transformation contexts:

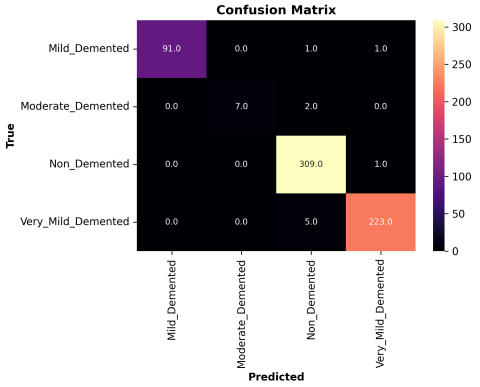
As for data transformation, it can be seen that this technique appears to have varying impacts depending on the model configuration. While it improves model generalization in some cases (notably seen in the slight enhancement with dropout and stride (2,2) under data transformation), it can also marginally reduce accuracy in others, as shown in Table I. This suggests that while generally beneficial for enhancing model robustness, data transformation must be carefully tuned based on the specific model architecture and configuration.

Regardless of data transformation presence, using stride (2,2) for subsampling tends to yield better results than max pooling. This indicates that stride (2,2) is more effective in preserving relevant information during dimensionality reduction compared to max pooling, which may lose significant details. Ajmera also highlighted in his article [15] that strided convolutions are particularly advantageous for applications requiring precise anomaly detection, such as medical imaging like MRI in this case. They offer improved performance in detecting minor anomalies and provide richer contextual information compared to max pooling.

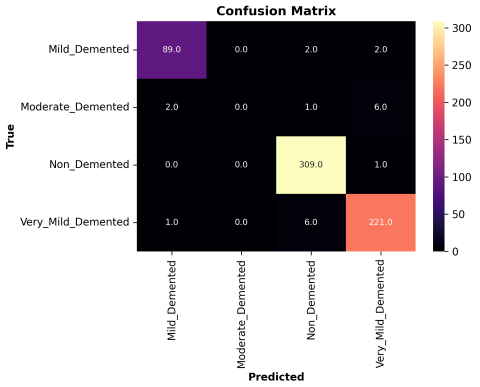
The effect of dropout varies based on the presence of data transformation and the subsampling method used. Without data transformation, dropout slightly improves accuracy when used with stride (2,2), suggesting its effectiveness in preventing overfitting in the absence of additional regularization techniques. Conversely, when data transformation is applied, the impact of dropout on model performance becomes less pronounced. This observation hints that data transformation techniques may alter the training data characteristics in ways that affect how dropout interacts with the model's learning dynamics, diminishing its relative contribution to preventing overfitting.

It can be observed that the presence of data transformation does not always guarantee accuracy improvement. This is evident from comparisons between configurations with and without data transformation, where sometimes the absence of data transformation results in slightly higher accuracy. This indicates that the effectiveness of data transformation heavily depends on the specific context, including dataset nature and model configuration.

In conclusion, the optimal choice between stride and max pooling, as well as the decision to include dropout, should be guided by a careful evaluation of the specific context, including the application of transformations to the data. While transformations applied to the data may improve the generalisation of the model, their impact varies depending on other architectural choices. Therefore, it is crucial to experiment with different combinations of these techniques to identify the configuration that maximises the model's accuracy on the specific dataset.



(a) Confusion matrix for the best model in presence of Data transformation, with Stride(2,2) and No Drop Out



(b) Confusion matrix for the worst model, **without** Data transformation, with MaxPooling and Drop Out

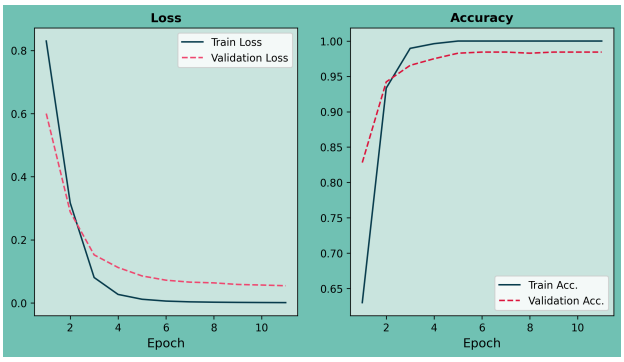
Fig. 4: Confusion matrix for the best and worst model for each configuration

V. Conclusion

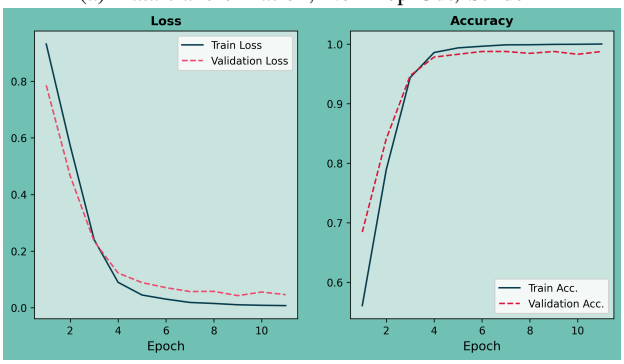
The experiments conducted in this study tested various configurations involving data transformation, Dropout, Stride, and MaxPooling to optimize the performance of a CNN model on a dataset of MRI images. The dataset posed challenges due to class imbalance, which was addressed using a class weighting technique.

Among the configurations tested, the model without data transformation, with Dropout, and Stride 2 achieved the highest overall accuracy on the test set, reaching 98.25%. This result is reasonable, since it suggests that the model performs better on non-transformed data because the images are in ideal conditions, free from blur, rotations and other distortions introduced by data transformation. Conversely, the model with data transformation, no Dropout, and Stride 2 achieved the highest accuracy among the transformed data configurations, reaching 97.50%. These results highlight that the model learns more effectively when trained on the original dataset, suggesting that data transformation can introduce variability that affects model performance. This underscores the importance of careful consideration in applying data preprocessing techniques.

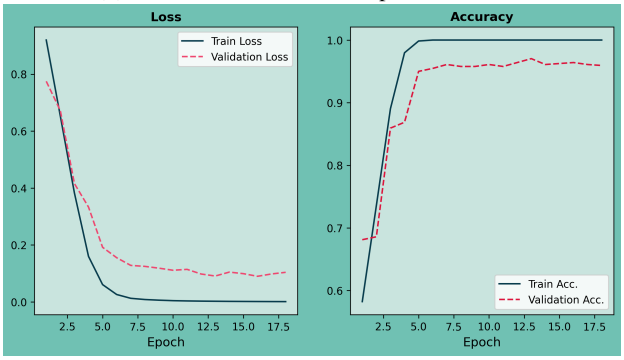
The overall accuracy achieved indicates the model's capa-



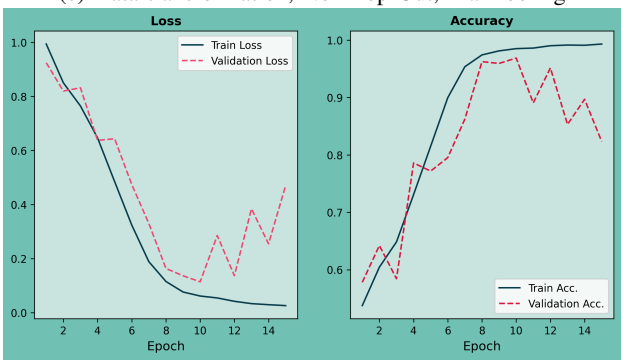
(a) Data transformation, No Drop Out, Stride 2



(b) Data transformation, Drop Out, Stride 2

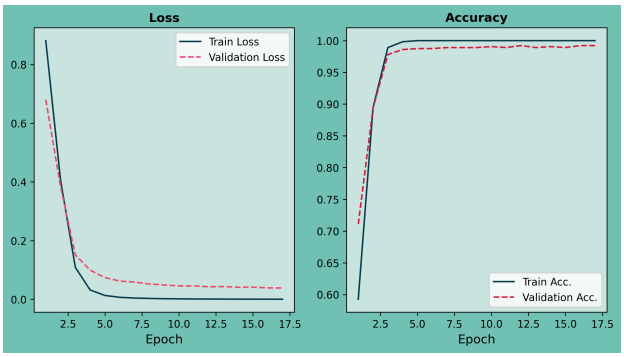


(c) Data transformation, No Drop Out, MaxPooling

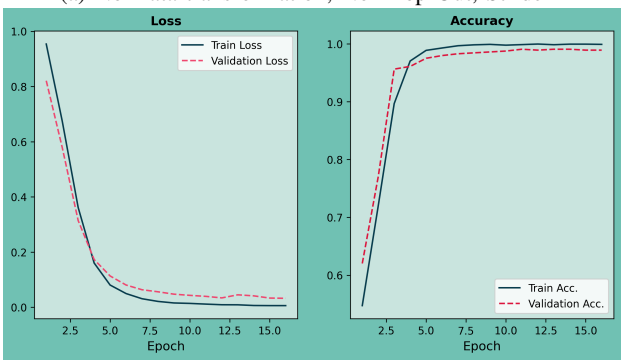


(d) Data transformation, Drop Out, MaxPooling

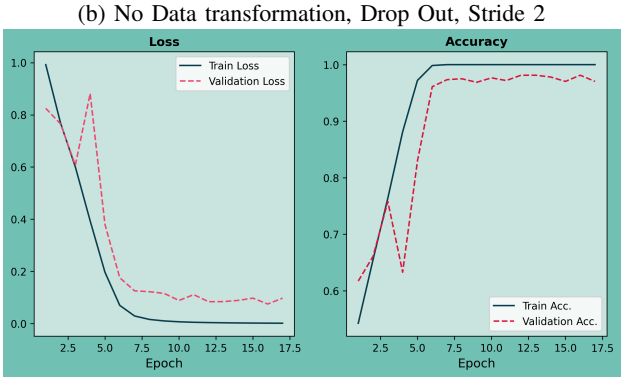
Fig. 5: Loss and Accuracy trends for models with Data transformation



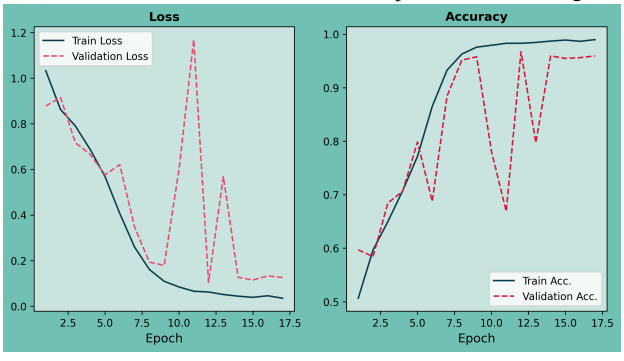
(a) No Data transformation, No Drop Out, Stride 2



(b) No Data transformation, Drop Out, Stride 2



(c) No Data transformation, No Drop Out, MaxPooling



(d) No Data transformation, Drop Out, MaxPooling

Fig. 6: Loss and Accuracy trends for models without Data transformation

bility to learn from the training data and make accurate predictions on unseen data. However, the presence of class imbalance and the differences in performance across configurations suggest that further improvements are necessary to achieve more consistent results. To address this issue, techniques such as Synthetic Minority Over-sampling Technique (*SMOTE*) could be employed in future work.

A possible future direction to enhance the model's robustness and accuracy involves testing it on additional datasets, such as the ADNI dataset, to gain insights into its generalizability and robustness across different data sources.

Moreover, although transfer learning was not utilized in this project due to the domain difference between pre-trained models on ImageNet and the MRI dataset used here, incorporating pre-trained models and fine-tuning their convolutional layers could be considered for future development. This approach could leverage the rich feature representations learned from large-scale datasets, potentially improving performance, as discussed in related works in Section II.

By addressing these aspects, future work can build upon the current model, enhancing its performance and making it a more reliable tool for classification tasks based on MRI images.

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